Zero Sum Subarray

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The problem

- Does any non-empty contiguous subarray sum to zero?
- A repeated prefix sum proves it.

Intuition

- If two prefix sums are equal, the elements between them sum to zero.

Approach

- 1. Keep a set of prefix sums starting with zero
- 2. Add each element to the running sum
- 3. If the sum was seen before, return true

Java solution

```java
public class ZeroSumSubarray {
    public static boolean zeroSumSubarray(int[] nums) {
        Set<Long> seen = new HashSet<>();
        seen.add(0L);
        long sum = 0;
        for (int n : nums) { sum += n; if (!seen.add(sum)) return true; }
        return false;
    }
}
```

Complexity

- Time: O(n)
- Space: O(n)

Recap

- If two prefix sums are equal, the elements between them sum to zero.
- Complexity: O(n)